

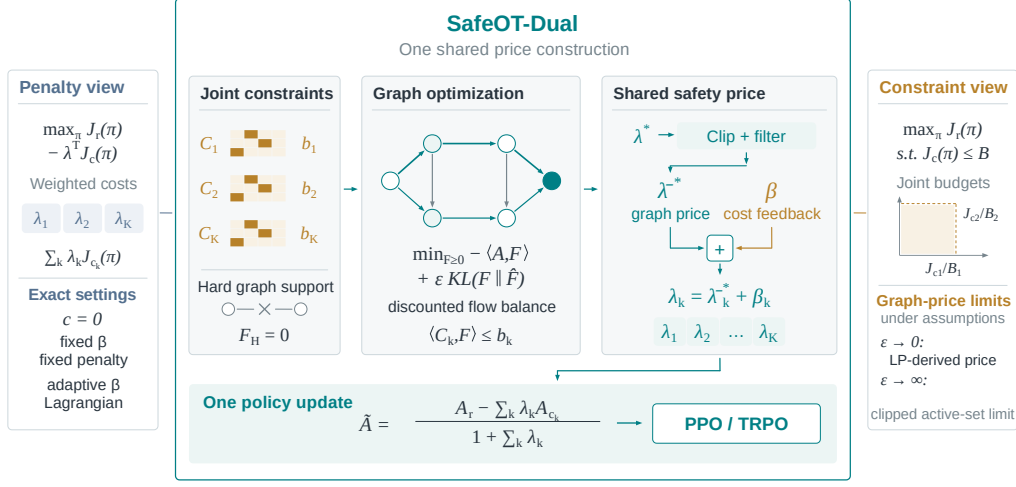


Figure 1: **One safety-price interface.** A clipped, filtered graph price is added to realized-cost feedback before pricing the policy advantages. At zero graph cap, fixed feedback gives a fixed penalty and adaptive feedback gives a Lagrangian rule. The graph LP and active-set limits require the assumptions in Section 3; they do not recover external algorithms such as CPO. Graphs are schematic.

Contributions.

- **A conditional analysis of graph prices.** We characterize the entropic projection's dual and end-point limits, including the assumptions under which clipping yields an active-set rule (Section 3).
- **A unified interface to policy learning.** Fixed penalties, feedback and graph-derived prices enter one cost-advantage surrogate; the algorithm separates this explicit transfer from graph feasibility and deployment safety (Section 4).
- **Mechanism controls.** Ten-task, five-seed comparisons against a realized-cost switch, warm-started feedback and PID show no consistent extra graph-solve benefit. Endpoint sweeps, integrator ablations, hard/soft tasks retain their protocol boundaries (Section 5).

2 PRELIMINARIES

CMDP occupancies. A constrained Markov decision process (Altman, 1999) is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \{c_k\}_{k=1}^K, \gamma, \mu_0)$ with transition kernel P , reward r , costs $c_k \geq 0$, discount $\gamma \in (0, 1)$ and initial distribution μ_0 . For a stationary policy π the normalised discounted occupancy and the value of any state-action function f are

$$\rho^\pi(s, a) = (1 - \gamma) \sum_{t \geq 0} \gamma^t \Pr_\pi(s_t = s, a_t = a), \quad J_f(\pi) = \sum_{s, a} \rho^\pi(s, a) f(s, a),$$

so reward and every cost are linear functionals of one object. With budgets $b_k \geq 0$ the CMDP is $\max_\pi J_r(\pi)$ s.t. $J_{c_k}(\pi) \leq b_k$, and the occupancy-measure theorem (Altman, 1999; Puterman, 1994) rewrites it as the linear program

$$\max_{\rho \geq 0} \sum_{s, a} \rho(s, a) r(s, a) \quad \text{s.t.} \quad \sum_a \rho(s', a) = (1 - \gamma) \mu_0(s') + \gamma \sum_{s, a} P(s'|s, a) \rho(s, a) \quad \forall s', \quad (1)$$

$$\sum_{s, a} \rho(s, a) c_k(s, a) \leq b_k \quad \forall k. \quad (2)$$

Eq. (1) characterises exactly the occupancies of stationary policies (recovered on visited states by $\pi(a|s) \propto \rho(s, a)$); Eq. (2) is a half-space per budget. Safety here is at the level of expectations; hard (never-to-be-incurred) costs are added in Section 3 as support constraints on the same object.

Occupancies as flows. Aggregating through the dynamics, $F_{ss'} = \sum_a \rho(s, a) P(s'|s, a)$ is the discounted mass transported along the edge $s \rightarrow s'$, and Eq. (1) becomes the discounted conservation

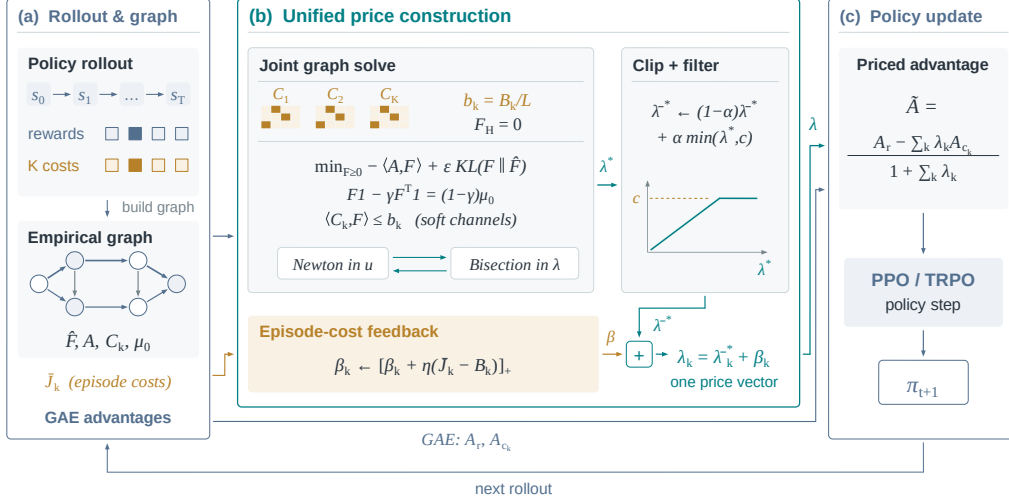


Figure 2: **From constraints to the policy update.** Rollouts supply an empirical graph, GAE advantages and episode-cost feedback. Graph budgets use $b_k = B_k/L$; feedback uses B_k . Only the solved prices cross from the graph to the policy surrogate, after clipping and filtering. The next policy supplies the next rollout. Graph feasibility alone does not certify policy safety; the separate off-policy executor is not shown.

Algorithm 1 SafeOT-Dual (on-policy instantiation)

- 1: **Input:** budgets B_k , typed channels ($\mathcal{K}_s$ soft, hard set), ε , nodes N , clip $\bar{\lambda}_{\max}$, integrator rate η , margin d
- 2: **for** each epoch t **do**
- 3: collect T steps with π_θ ; hard-channel training cost $\leftarrow$ contact $\vee$ (lidar within d)
- 4: compute GAE advantages A_r, A_{c_k} ; discretize states (k-means, warm-started refits keep node ids)
- 5: build $\hat{F}, A, C_k$ from the rollout (rolling window for $\hat{F}, C_k$); $b_k \leftarrow B_k/L$
- 6: solve (3) by damped Newton in u and coordinate bisection in $\lambda \Rightarrow \lambda^*$ (numerical solver; residual checks in Appendix A)
- 7: $\bar{\lambda}_k^* \leftarrow (1 - \alpha)\bar{\lambda}_k^* + \alpha \min(\lambda_k^*, \bar{\lambda}_{\max})$; $\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+$; $\lambda_k \leftarrow \bar{\lambda}_k^* + \beta_k$ (hard channels: $\lambda_{\max}$ if zero-capacity)
- 8: policy step on $(A_r - \sum_k \lambda_k A_{c_k}) / (1 + \sum_k \lambda_k)$: PPO clip with target-KL early stopping, or one TRPO natural-gradient step with line search
- 9: **end for**

4 SAFEOT-DUAL

SafeOT-Dual replaces the multiplier source in PPO-Lagrangian or TRPO-Lagrangian with the graph price plus feedback (5). Algorithm 1 specifies this on-policy interface. The separate off-policy instantiation prices the soft channel in a SAC-style update and assigns the hard channel to an FDPI feasibility critic (Yang et al., 2026) (Appendix B).

What reaches the actor. For each rollout sample, the policy optimizer receives

$$\tilde{A}_t = \frac{A_{r,t} - \sum_k \lambda_k A_{c_k,t}}{1 + \sum_k \lambda_k}, \quad \lambda_k = \bar{\lambda}_k^* + \beta_k. \quad (6)$$

GAE supplies the reward and cost advantages; the graph supplies their coefficients. Positive prices insert cost-advantage terms directly into the surrogate. Finite optimization, clipping and policy-class restrictions determine the realized update, so this interface does not imply final-policy feasibility.